

Kenar Narayaka

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EDUCATION

Univeresity of Oxford – Christ Church

Oxford, England

MCompPhil Computer Science & Philosophy - Predicted 1st Class Honours

2025–2029

Relevant Coursework: Probability, Linear Algebra, Logic, Discrete Maths, Design & Analysis of Algorithms, Algorithms & Data Structures, Continuous Maths, Functional Programming, Imperative Programming, Logic and Proof, Models of Computation

Loughborough College

Loughborough, England

A-Level Maths, Further Maths, Computer Science - A*AA

2023–2025

EXPERIENCE

Hitachi Astemo

Jun. 2024 - Aug. 2024

Backend Engineer Intern - Human Resources | PHP, Fat-Free Framework (F3), Postgres, Docker

- Replaced paper-based HR processes covering ~15,000 employee records with administrative tools, translating legacy forms, approval states, and organisational relationships into normalised Postgres schemas.
- Designed and Dockerised REST APIs and a multi-stage approval engine with server-side validation, RBAC, immutable audit history, and fulfilment tracking; reduced end-to-end request turnaround from ~7 days to 5 hours (-97%)
- Implemented permission-scoped CRUD dashboard and coordinate-based PDF generation connected to network printers, then extracted form components and print-template logic into reusable tooling for later workflows.

CapitOx

Jan. 2026 - Apr. 2026

Quant Academy – Oxford-Man Institute | C++, Python, NumPy, SciPy, Matplotlib

- Selected for a competitive programme covering mean-variance portfolio construction, factor models, signal combination, and the design and evaluation of multi-signal systematic strategies
- Built a Monte Carlo pricing pipeline for a European put under a 2-component generalised-normal mixture model, stable log-density evaluation and separating target distribution, proposal mechanism, and estimator logic for controlled comparison.
- Implemented 3 estimators with proposal tuning, convergence diagnostics, and seeded experiments; evaluated across 100+ trials and achieved 5x lower variance and 60x lower runtime at matched error relative to baseline random-walk Metropolis-Hastings.

PROJECTS

Fracta - 3D Fractal Simulator

 | JavaScript, three.js, GLSL

- Built a real-time browser-based render for fractal families, combining signed-distance estimation with custom GLSL raymarching, configurable lighting and colour pipelines, and camera controls exposed by a web interface.
- Implemented distance-adaptive step sizing that used estimated scene geometry to reduce unnecessary samples and increase frame rate by up to 40%, then added for a further ~20% improvement over the original pipeline.
- Profiled performance across camera distances, iteration counts, and fractal params to balance throughput against close-range geometric detail; awarded 69/70, highest OCR Computer Science NEA score in school history.

shotLab - Basketball Jumpshot Analyser

 | Tauri v2, Rust, ONNX Runtime, FFmpeg

- Built a Tauri v2 + Svelte app that analyses basketball jumpshot form from a single video, designed behind a Rust analysis core so all processing is done locally; ran top-down RTMPose estimation through ONNX Runtime with SimCC keypoint decoder.
- Decoded footage through FFmpeg keyed on trueframe timestamps to stay accurate on variable frame rates, then segmented continuous multi-shot recordings into per-rep phases and extracted 28 biomechanical parameters across 6 kinematic families.
- Ranked coaching cues by shot-to-shot consistency behind a typed availability model and fit a gravity-constrained release parabola with closed-form least squares to estimate arc consistency by aligning each rep to a proportion-scaled ideal model via dynamic time-warping

TECHNICAL SKILLS

Languages: Python, C++, PHP, JavaScript, SQL, GLSL, Rust

Frameworks & Libraries: NumPy, SciPy, Matplotlib, three.js, Fat-Free Framework, Tauri V2

Tools: Postgres, Git, Linux, tmux, Docker

Concepts: Backend Development, Database Design, Graphics Programming, REST APIs